



VoxelNet

End-to-End Learning for Point Cloud Based 3D Object Detection

Zhou, Yin, and Oncel Tuzel. Proceedings of the IEEE conference on computer vision and pattern recognition. (2018)

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1. Introduction

What is 3D Object Detection?

Problem definition

$$\mathcal{B} = f_{det}(\mathcal{I}_{sensor}),$$

$\mathcal{B} = \{B_1, \dots, B_N\}$ is a set of N 3D object in a scene,

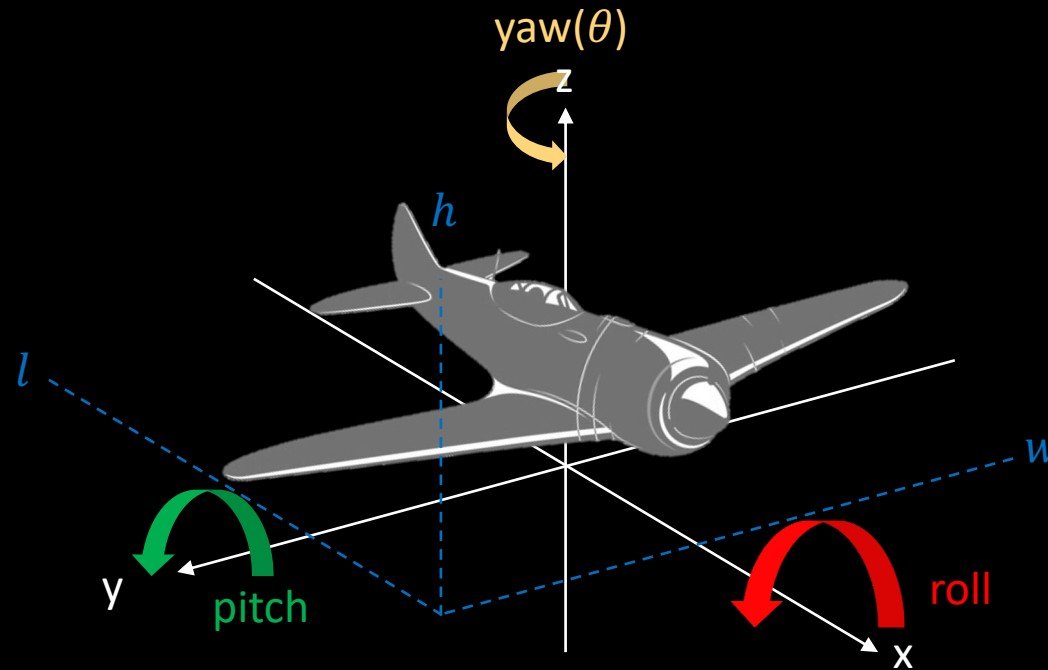
f_{det} is a 3D object detection model,

$\mathcal{I}_{sensor}$ is one or more sensory inputs.

3D Cuboid

$$B = [x_c, y_c, z_c, l, w, h, \theta, class]$$

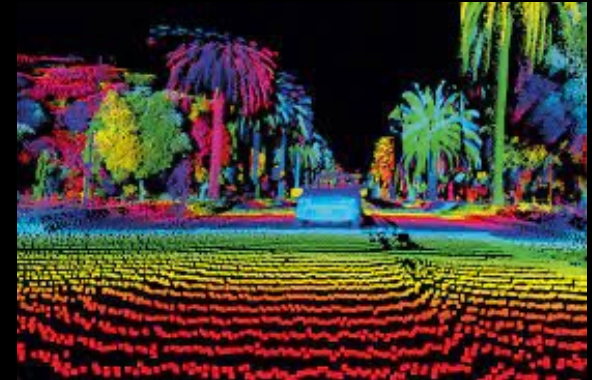
v_x, v_y - speed



Sensory Inputs

Radars, cameras, and LiDAR (Light Detection And Ranging) sensors are the three most widely adopted sensory types

- Radar: Long detection range and robust to weather conditions. Velocity(Doppler)
- Camera: Cheap and easily accessible and crucial for understanding semantics.
- LiDAR: Accurate 3D information directly acquired by LiDAR sensors.



Comparisons with 2D Object Detection

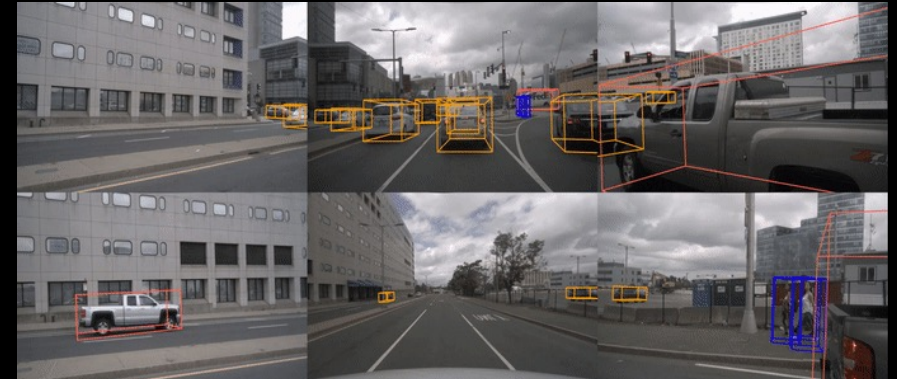
- Heterogeneous data representations.
- 2D methods detect from the perspective view. 3D methods must consider different views.
- 3D methods has a high demand for accurate localization in the 3D space.



Bird's Eye View(LiDAR)



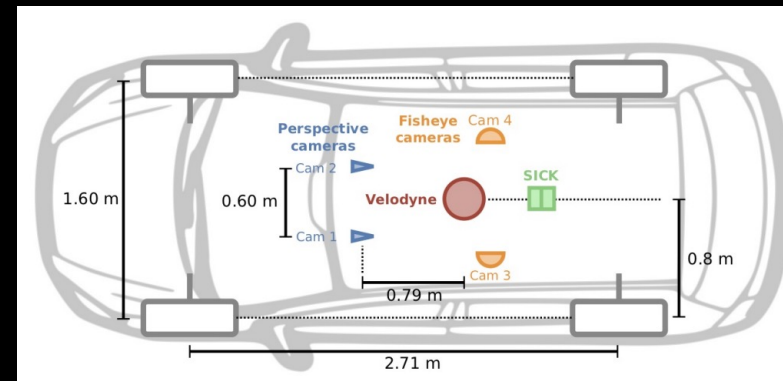
Point View



Cylindrical View

Datasets - KITTI

- KITTI: Pioneering work data collection and annotating 3D objects from the collected data.
- 3D IoU



Geiger, A., Lenz, P., Stiller, C., & Urtasun, R. (2013).
Vision meets robotics: The kitti dataset.
The International Journal of Robotics Research, 32(11), 1231-1237.

VoxelNet

- Voxel feature encoding (**VFE**) layer, which enables **inter-point interaction**.
- Stacking multiple VFE layers allows learning **complex feature**.
- VoxelNet divides the point cloud into equally spaced **3D voxels**, encodes each voxel via stacked **VFE layers**, and then **3D convolution** further aggregates local voxel features, transforming the point cloud into a high-dimensional volumetric representation and yield the detection result.

→ Benefits both from the **sparse point** structure and **parallel processing** on the voxel grid.

2. Architecture

$$Z \times Y \times X = [-3, 1] \times [-40, 40] \times [0, 70.4]$$
$$D, H, W = 10, 400, 352$$

Feature learning network

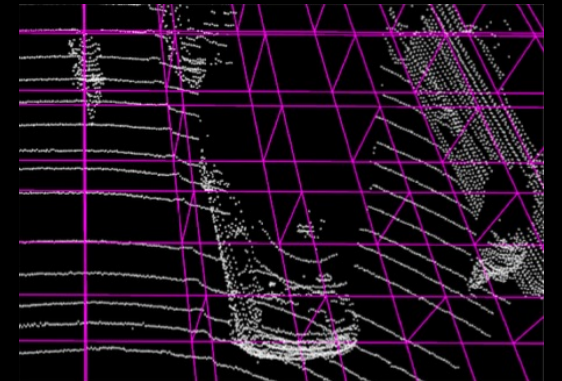
Voxel Partition

- Subdivide the 3D space into equally spaced voxels.
- Suppose the point encompasses with range D, H, W along the Z, Y, X axes respectively.

voxel of size $v_D, v_H, v_W = 0.4, 0.2, 0.2$

D, H, W are multiple of v_D, v_H, v_W

$D, H, W = Z, Y, X$
 $H, W, L = Z, Y, X$



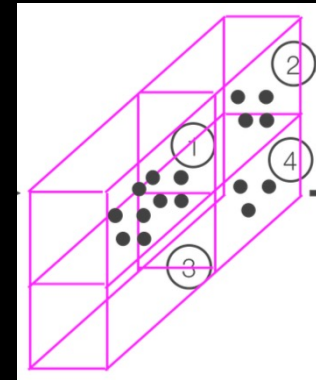
Feature learning network

Grouping

- LiDAR point cloud is sparse and has highly variable point.
- Therefore, after grouping, a voxel will contain a variable number of points.

Random Sampling

1. Computational savings
2. Decreases the imbalance



Feature learning network

Stacked Voxel Feature Encoding

- $V = \{p_i = [x_i, y_i, z_i, r_i]^T \in \mathbb{R}^4\}_{i=1\dots t}$ as a non-empty voxel containing $t \leq T$ LiDAR points, where p_i contains XYZ coordinates for the i -th point and r_i is the received reflectance.
- Local mean as the centroid of all the points in $V(v_x, v_y, v_z)$
- Augment each point p_i $V_{in} = \{\hat{p}_i[x_i, y_i, z_i, r_i, x_i - v_x, y_i - v_y, z_i - v_z]^T \in \mathbb{R}^7\}_{i=1\dots t}$ transformed through the fully connected network (FCN) into a feature space

Sparse Tensor Representation

$$4D = C \times D' \times H' \times W' = 128 \times 10 \times 400 \times 352$$

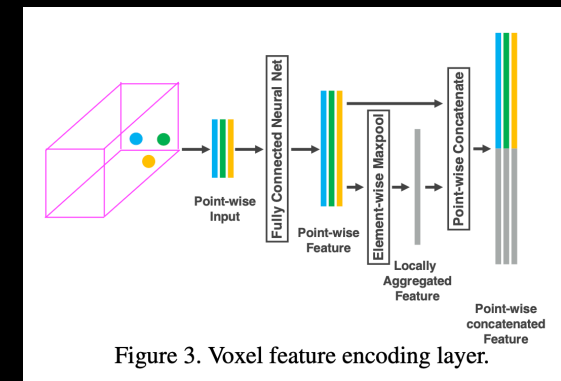
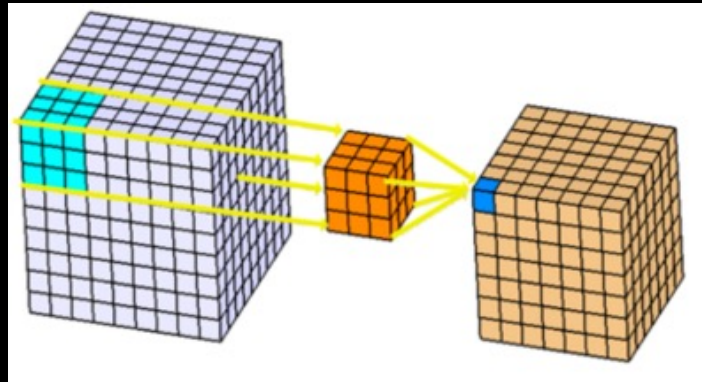


Figure 3. Voxel feature encoding layer.

Convolutional Middle Layers

- $\text{ConvMD}(c_{\text{in}}, c_{\text{out}}, k, s, p)$ to represent an M -dimensional convolution operator where c_{in} and c_{out} , kernel size(k), stride size(s) and padding size(p).

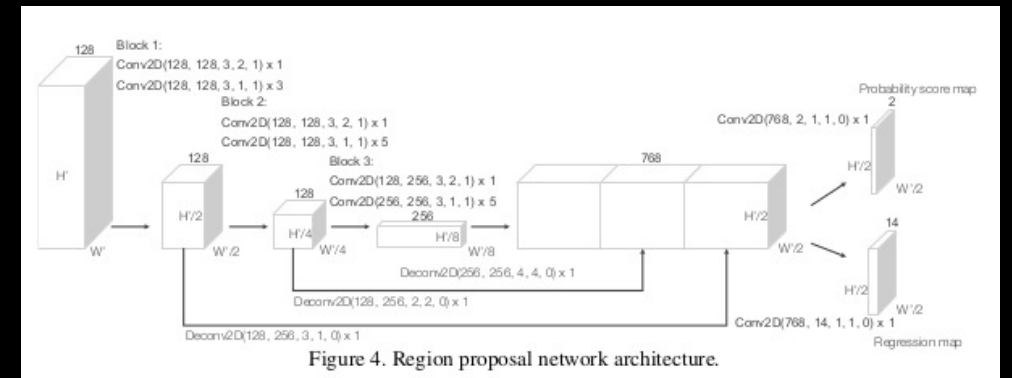
$$4D = C \times D' \times H' \times W' = 64 \times 2 \times 400 \times 352$$



Region Proposal Network

The network has three blocks of fully convolutional layers. The first layer of each block downsamples the feature map by half via a convolution with a stride size of 2, followed by a stride 1. BN, ReLU. Upsample the output of every block a fixed size and concatenate to construct the high resolution feature map. 1. score map, 2. regression map

$$3D = C \times H' \times W' = 128 \times 400 \times 352$$



Loss Function

$$L = \alpha \frac{1}{N_{pos}} \sum_i L_{cls}(p_i^{pos}, 1) + \beta \frac{1}{N_{neg}} \sum_j L_{cls}(p_j^{neg}, 0) \\ + \frac{1}{N_{pos}} \sum_i L_{reg}(u_i, u_i^*)$$

Let $\{a_i^{pos}\}_{i=1\dots N_{pos}}$ be the set of N_{pos} positive anchors
 $\{a_j^{neg}\}_{j=1\dots N_{neg}}$ be the set of N_{neg} negative anchors.

A 3D ground truth box as $(x_c^g, y_c^g, z_c^g, l^g, w^g, h^g, \theta^g)$, where x_c^g, y_c^g, z_c^g represent the center location, l^g, w^g, h^g are length, width, height of the box, and θ^g is the yaw rotation around Z-axis.

To retrieve the ground truth box from a matching positive anchor parameterized as $(x_c^a, y_c^a, z_c^a, l^a, w^a, h^a, \theta^a)$
 $u^* \in \mathbb{R}^7 = \Delta x, \Delta y, \Delta z, \Delta l, \Delta w, \Delta h, \Delta \theta$

3. Experiments

Evaluation in 3D

Method	Modality	Car			Pedestrian			Cyclist		
		Easy	Moderate	Hard	Easy	Moderate	Hard	Easy	Moderate	Hard
Mono3D	Mono	2.53	2.31	2.31	N/A	N/A	N/A	N/A	N/A	N/A
3DOP	Stereo	6.55	5.07	4.10	N/A	N/A	N/A	N/A	N/A	N/A
VeloFCN	LiDAR	15.20	13.66	15.98	N/A	N/A	N/A	N/A	N/A	N/A
MV (BV+FV)	LiDAR	71.19	56.60	55.30	N/A	N/A	N/A	N/A	N/A	N/A
MV (BV+FV+RGB)	LiDAR+Mono	71.29	62.68	56.56	N/A	N/A	N/A	N/A	N/A	N/A
HC-baseline	LiDAR	71.73	59.75	55.69	43.95	40.18	37.48	55.35	36.07	34.15
VoxelNet	LiDAR	81.97	65.46	62.85	57.86	53.42	48.87	67.17	47.65	45.11

Evaluation in 3D



4. Conclusion

Conclusion

- Remove the bottleneck of manual feature engineering and propose VoxelNet.
- Operate directly on sparse 3D points and capture 3D shape information effectively.
- Efficient implementation of VoxelNet that benefits from point cloud sparsity and parallel processing on a voxel grid.
- Show that VoxelNet outperforms state-of-the-art LiDAR based 3D detection methods by a large margin.
- Provides a better 3D representation.

Future work: Extending VoxelNet for joint LiDAR and image based end-to-end 3D detection to further improve detection and localization accuracy.